EUGENE LOTSU

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EDUCATION

William Jewell College **Liberty,Mo**

Bachelor of Science, Computer Science *Expected May '28*

GPA: *3.88*

Relevant Coursework: Introduction to Programming, Programming for Everyone II, Web Development, Data Structures, Data Visualization, Discrete Mathematics

SKILLS

Languages: Python, JavaScript, TypeScript, SQL
Frameworks / Tools: React, Node.js, Express, PostgreSQL, Git, AWS (Lambda, S3, DynamoDB, API Gateway, Cognito, Rekognition, Bedrock), Vercel, Postman
ML / AI: Computer Vision (Rekognition), LLM APIs (Claude via Bedrock), Prompt-based systems design
Concepts: REST APIs, Authentication systems, Serverless architecture, Data pipelines, Distributed systems basics, Algorithms, Data Structures
Organizations: ColorStack

PROJECTS & EXPERIENCES

Build4Me | AI Construction Verification Platform ***(Semifinalist, AWS 10,000 AIdeas Competition (AI Hackathon))***

- Designed and built a serverless AI verification system for diaspora-funded construction projects using AWS Lambda, API Gateway, DynamoDB, and S3
- Engineered an AI verification pipeline combining computer vision (Amazon Rekognition), LLM reasoning (Amazon Bedrock), and GPS validation (Amazon Location Service) to assess construction milestone authenticity
- Built full-stack interface (Next.js + TypeScript) with role-based dashboards for contractors, project owners, and inspectors
- Deployed secure system using Cognito authentication, JWT authorization, and pre-signed S3 uploads for scalable media handling

Configurable Automation Agent (Python, Gemini)

- Built a modular Python agent system capable of multi-step reasoning and iterative task execution with persistent state tracking
- Designed a controlled execution sandbox enabling safe file system reads/writes and structured tool usage
- Implemented loop-based orchestration engine for dynamic planning, execution, and self-correction of tasks
- Applied system to CLI modification and program verification tasks, demonstrating automated code analysis and transformation

JWT Authentication API | Node.js, Express, MongoDB

- Built secure authentication system with user registration, login, and protected routes using JWT and bcrypt hashing
- Designed RESTful API architecture with Express and Mongoose-based data modeling in MongoDB
- Implemented middleware-based authorization layer for route protection and session validation
- Tested and validated endpoints using Postman with edge case handling for authentication flows

Build4Me API

Node.js, Express, PostgreSQL, Sequelize, JWT, Jest

- Architected and deployed a scalable backend for a construction milestone verification platform supporting multi-role workflows
- Designed relational data models and built RESTful APIs to manage project lifecycles, updates, and inspection-based payment gating
- Implemented secure authentication and role-based authorization using JWT and bcrypt for protected access control
- Improved reliability through automated API testing (Jest, Supertest) and structured documentation for developer onboarding

WORK & LEADERSHIP EXPERIENCE

Vice President, African Student Union **Aug 20, 2025–** ***Dec 15,2025***

William Jewell College

- Coordinated cultural and educational campus events promoting African heritage and student engagement
- Led executive board planning sessions and supported strategic event execution
- Managed committees, budgets, and partnerships to expand organization participation

Data Analyst Associate Intern| Project Lead **Aug 4, 2025-Sept 4,2025**

Excelerate(in collaboration with Saint Louis University)

- Led cross-functional analytics project analyzing multi-source datasets using Python (Pandas) and SQL
- Built interactive dashboards in Looker Studio and Seaborn to identify KPI trends and business insights
- Coordinated weekly deliverables across a distributed global team, ensuring on-time analytical output
- Translated raw data into actionable recommendations for academic and institutional stakeholders

AWARDS & INVOLVEMENT

ColorStack,*Member* **September 2024-Present**

Semifinalist, AWS 10,000 AIdeas Competition (Top 1000 applicants; awarded AWS credits)